

Csuprem Tutorial *(3D)*



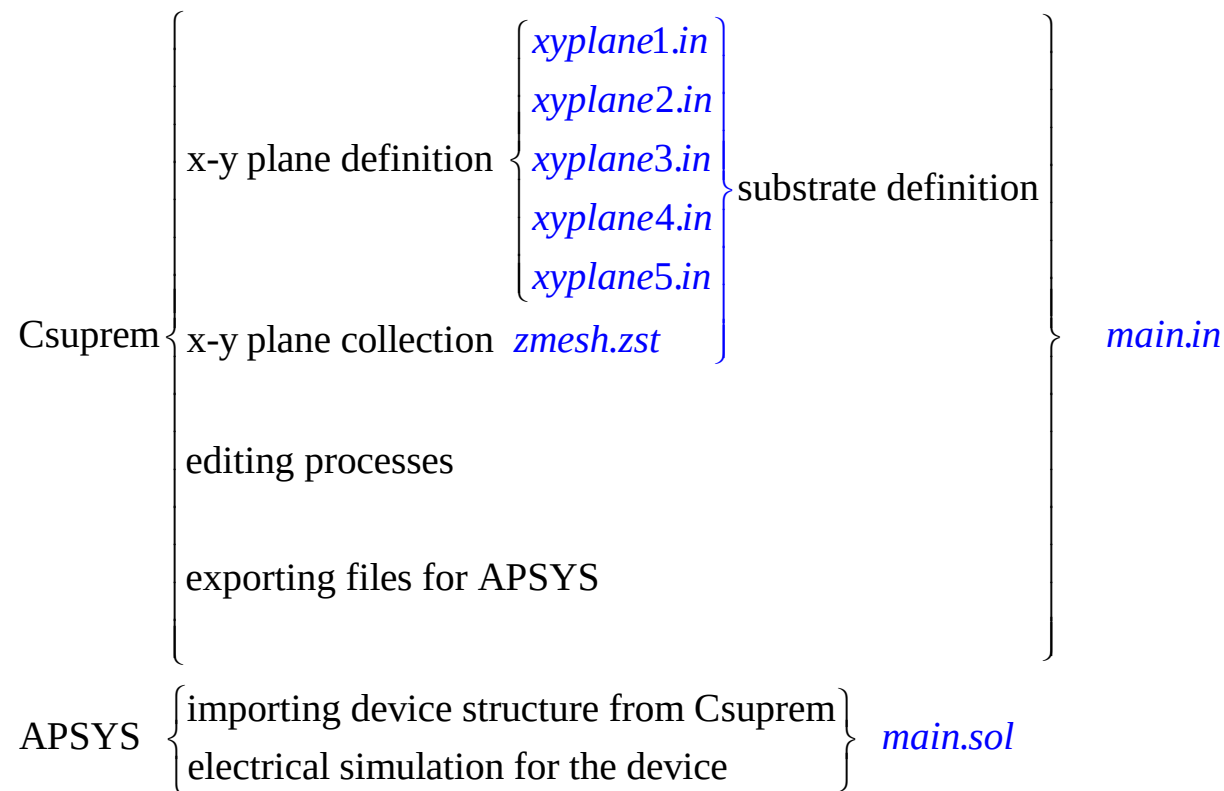
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Outline for a whole course of processing and device simulation

- Build a 3D substrate in Csuprem software
- Simplified processes for nMOSFET
- Graphical demonstration of 3D process simulation
- 3D device simulation by Apsys software

We firstly show file structure in this tutorial





Build a 3D structure

- ① Set up x-y planes, each x-y plane is defined in an individual file.
- ② specify positions of these parallel planes in z direction, which is stored in a file with a fixed name *zmesh.zst*.
- ③ Specify dimensionality(3D) in main input.
- ④ Initialize 3D mesh.

Build a 3D structure

Mesh define

A line starting from “#” is a line of comment

line x loc=0.0 tag=lft spacing=0.04

line x loc=@gatelen/2 spacing=0.02

line x loc=@gatelen/2+0.2 tag=rht spacing=0.04

line y loc=0.0 tag=top spacing=0.01

line y loc=0.1 spacing=0.03

line y loc=0.5 spacing=0.05

line y loc=1.0 spacing=0.1

line y loc=2. tag=bot spacing=0.3

“line” command is used to specify a flag point in x or y direction. The point can have a “tag” to be cited in other commands. Between these flag points, mesh points will be inserted with density controlled by “spacing”. Smaller “spacing” means more mesh points. In addition, “@gatelen” is a variable whose value must be specified before its citation.

After flag points in x and y direction are defined, they will form a rectangle 2D mesh like a 2D matrix where each mesh point acts as a matrix element.

This is content of **xyplane1.in** which is same as **xyplane2.in**, **xyplane3.in**, **xyplane4.in** and **xyplane5.in**

Build a 3D structure

```
elim y.dir xlo=0 xhi=@gatelen/2+0.2 ylo=0.5 yhi=2 ntimes=2
```

```
elim x.dir xlo=0 xhi=@gatelen/2.5 ylo=0 yhi=2 ntimes=1
```

“elim” command is used to eliminate mesh points in x or y direction (x.dir or y.dir). These mesh points, belonging to a rectangular area specified by “xlo”, “xhi”, “ylo” and “yhi”, will be halved “ntimes” times.

```
region silicon xlo=lft xhi=rht ylo=top yhi=bot
```

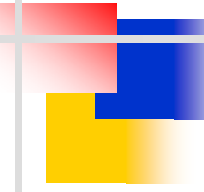
```
bound exposed xlo=lft xhi=rht ylo=top yhi=top
```

```
bound backside xlo=lft xhi=rht ylo=bot yhi=bot
```

“region” command specifies a type of material(silicon here) in a rectangular region defined by “xlo”, “xhi”, “ylo” and “yhi”, here “tag” of flag point is used.

“bound” command defines “exposed” or “backside” area for the substrate.

This is content of xyplane1.in which is same as xyplane2.in, xyplane3.in, xyplane4.in and xyplane5.in



Build a 3D structure

- ① Set up x-yplanes, each x-y plane is defined in an individual file.
- ② specify positions of these parallel planes in z direction, which is stored in a file with a fixed name *zmesh.zst*.
- ③ Specify dimensionality(3D) in main input.
- ④ Initialize 3D mesh.

Build a 3D structure

begin_zst

z_structure uniform_zseg_from=0.0 uniform_zseg_to=0.0 zplanes=1 zseg_num=1

z_structure uniform_zseg_from=0.333 uniform_zseg_to=0.333 zplanes=1 zseg_num=2

z_structure uniform_zseg_from=0.389 uniform_zseg_to=0.389 zplanes=1 zseg_num=3

z_structure uniform_zseg_from=0.444 uniform_zseg_to=0.444 zplanes=1 zseg_num=4

z_structure uniform_zseg_from=0.5 uniform_zseg_to=0.5 zplanes=1 zseg_num=5

z_structure uniform_zseg_from=0.501 uniform_zseg_to=0.501 zplanes=1 zseg_num=6

z_structure uniform_zseg_from=2.5 uniform_zseg_to=2.5 zplanes=1 zseg_num=7

“z_structure” command is used to specify position of a x-y plane and is supported by both Csuprem and APSYS softwares. x-y planes are numbered by “zseg_num” and located by “uniform_zseg_from” and “uniform_zseg_to”. “zplanes” tells how many times this x-y plane will be repeated before next x-y plane. But we fix “zplanes” equals 1 in Csuprem to make an uniform segment between 2 x-y planes, which helps preventing abnormal crash when doing simulation of “oxidation” processing.

This is content of zmesh.zst

Build a 3D structure

```
load_mesh mesh_inf=f1.msh zseg_num=1
load_mesh mesh_inf=f2.msh zseg_num=2
load_mesh mesh_inf=f3.msh zseg_num=3
load_mesh mesh_inf=f4.msh zseg_num=4
load_mesh mesh_inf=f5.msh zseg_num=5
load_mesh mesh_inf=f6.msh zseg_num=6
load_mesh mesh_inf=f7.msh zseg_num=7
```

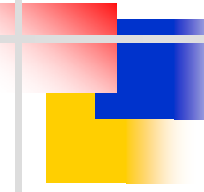
These “load_mesh” commands are formative and repeated by times of x-y plane amount with a simple rule of number increasing.

```
output sol_out=tmp.out
export_3dgeo file=h_cvd.3dgeo
```

These 2 lines are of fixed format, please do not modify them.

```
end_zst
```

This is content of **zmesh.zst**



Build a 3D structure

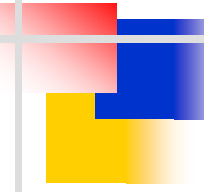
- ① Set up x-yplanes, each x-y plane is defined in an individual file.
- ② specify positions of these parallel planes in z direction, which is stored in a file with a fixed name *zmesh.zst*.
- ③ Specify dimensionality(3D) in main input.**
- ④ Initialize 3D mesh.

Build a 3D structure

mode three.dim

This command tells simulator the dimensionality and if to consider interaction between x-y planes(`three.dim`) or not to(`quasi3D`) for 3D simulation.

This is at head of `main.in` before processing simulation



Build a 3D structure

- ① Set up x-yplanes, each x-y plane is defined in an individual file.
- ② specify positions of these parallel planes in z direction, which is stored in a file with a fixed name *zmesh.zst*.
- ③ Specify dimensionality(3D) in main input.
- ④ Initialize 3D mesh.

Build a 3D structure

This is at head of **main.in** before processing simulation

```
3d_mesh nsegm=7 inf=mesh
```

“3d_mesh” command loads files which define x-y planes. File amount is specified by “nsegm”

```
init or=100
```

Finally, “init” command sets up a full substrate with all necessary information. “or” is used to specify the substrate orientation.

“zmesh.zst” will automatically cited by simulator, users do not need to evidently include it in “main.in”

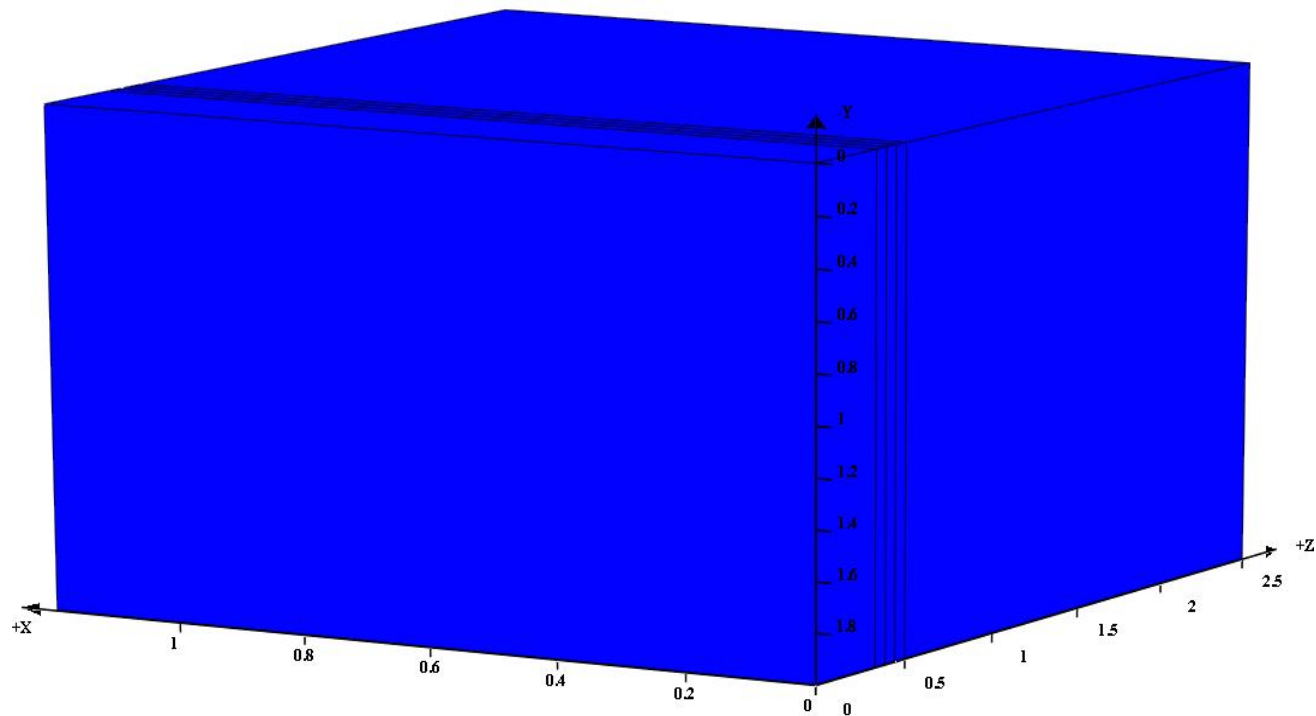
Simplified processes

Steps

1. 3D-STI
2. deposit gate oxide
3. channel implant
4. deposit gate poly
5. anneal after deposit
6. etch some poly
7. anneal after deposit
8. Idd implant for source/drain
9. make spacer
10. anneal in dry O2 atmosphere
11. final implant for source/drain
12. etch off oxide on source/drain region
13. cut off some extra mesh points to save run time for device simulation and left-mirror copy the structure to get a full MOSFET
14. export final structure to an APSYS-supported file

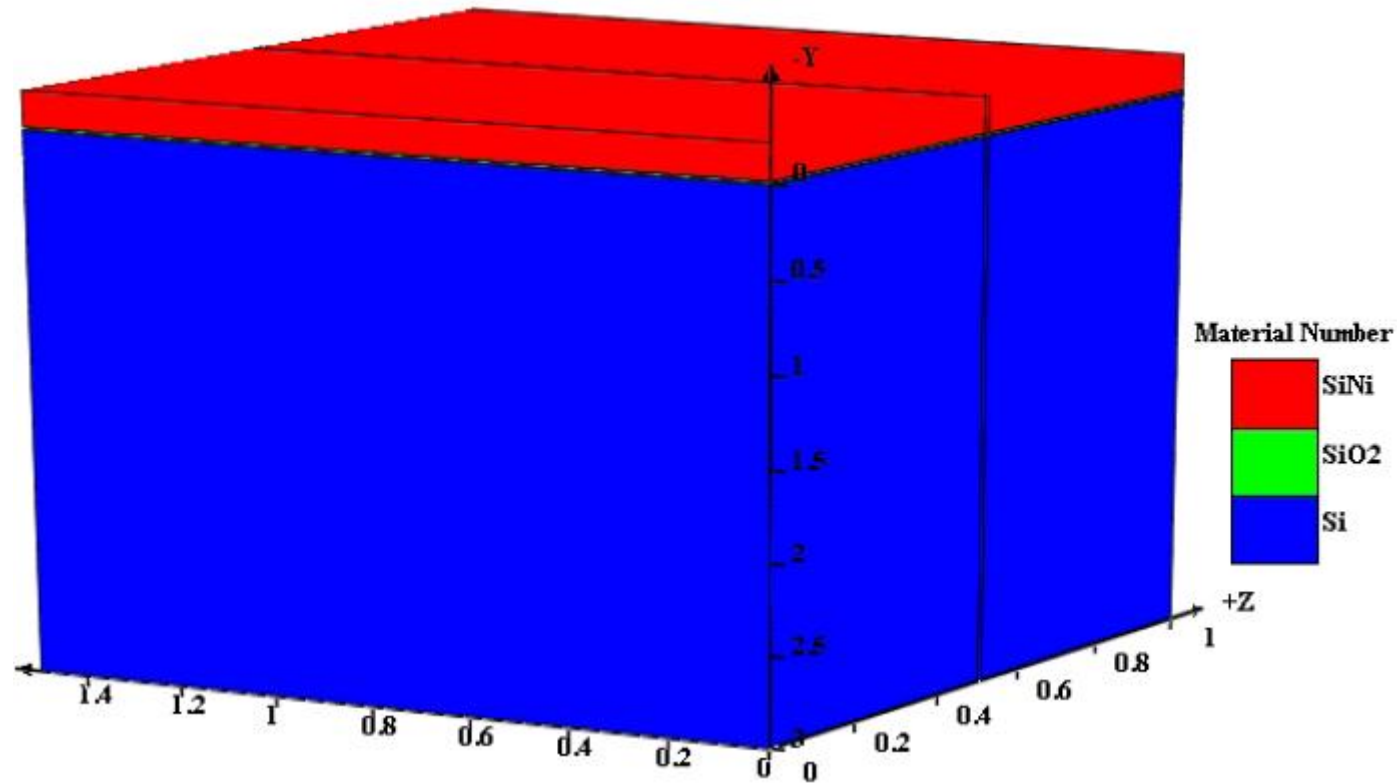
Graphical demonstration

Substrate



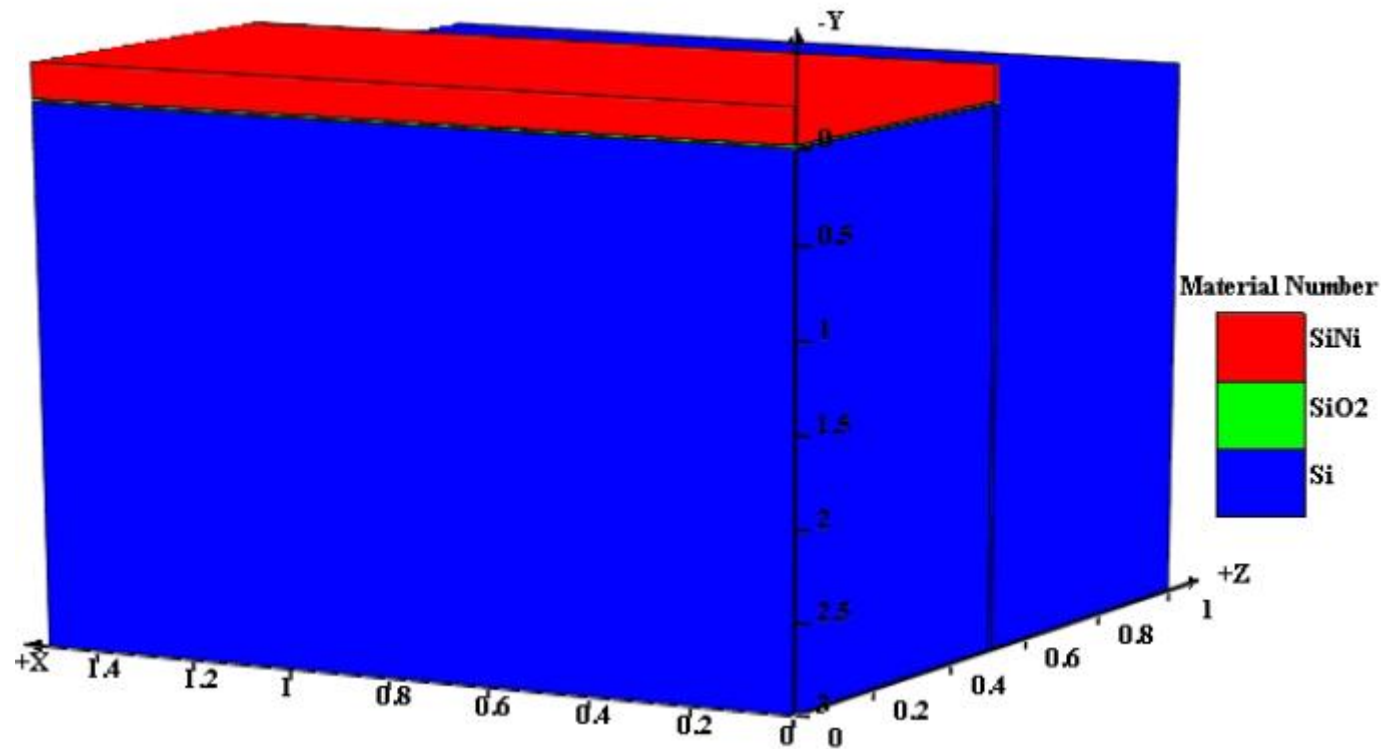
Graphical demonstration

Step 1 STI_01_cover



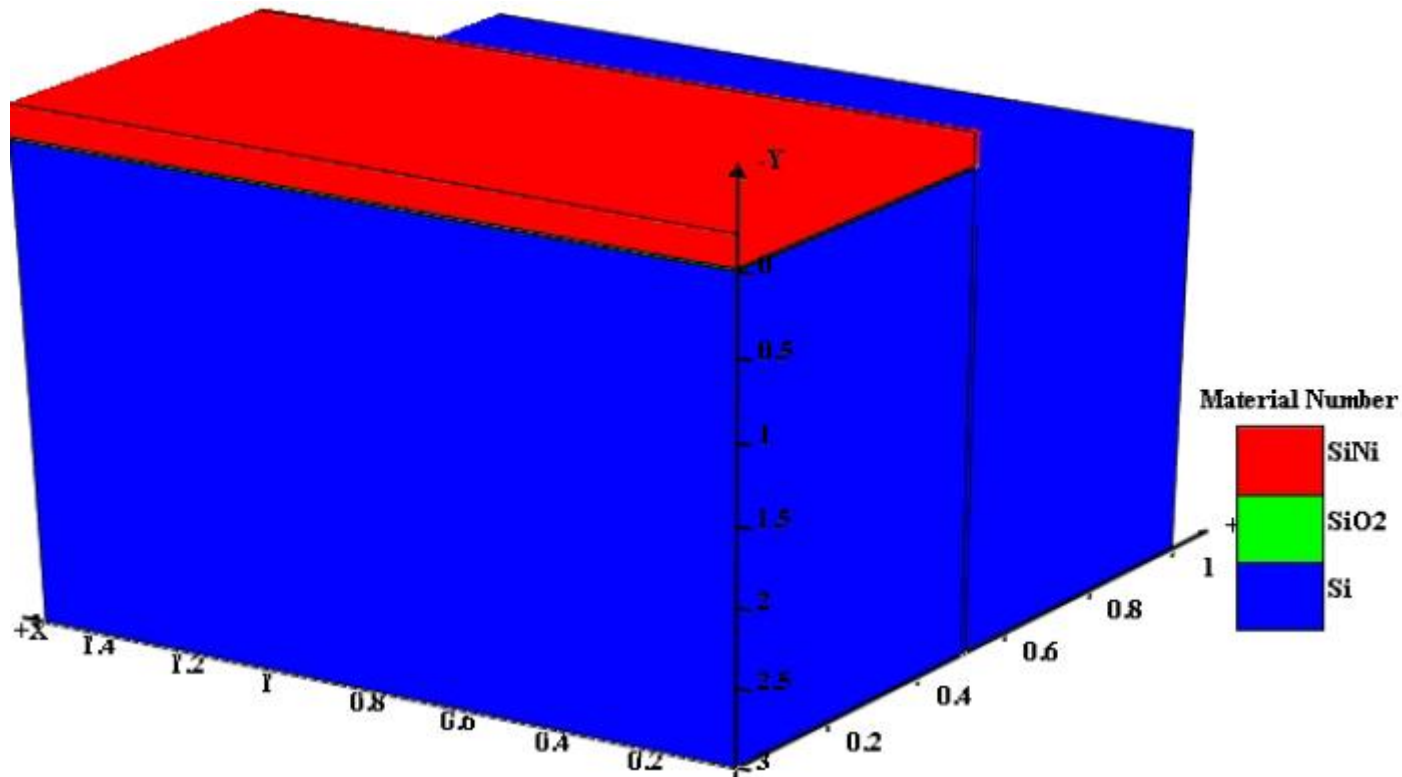
Graphical demonstration

Step 1 STI_02_decover



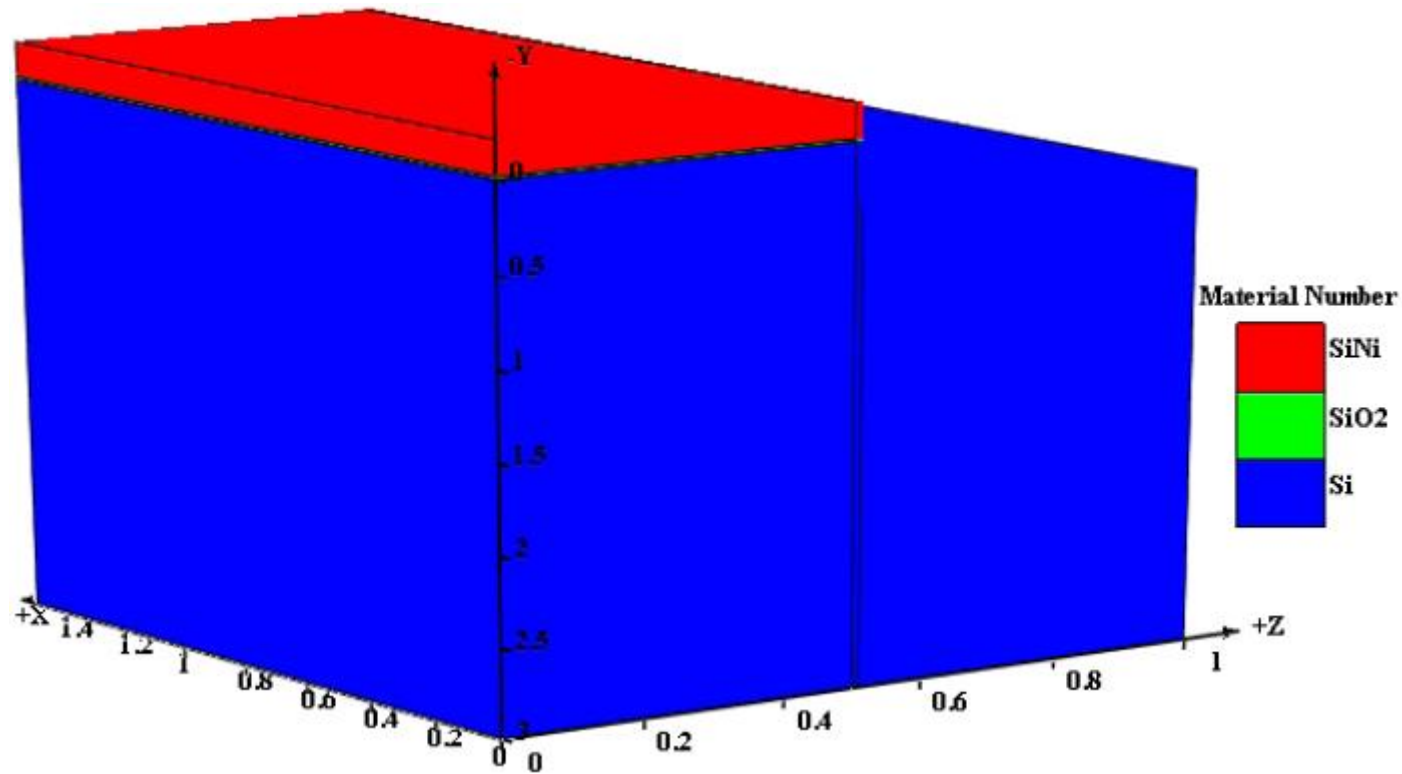
Graphical demonstration

Step 1 STI_03_etch_trench



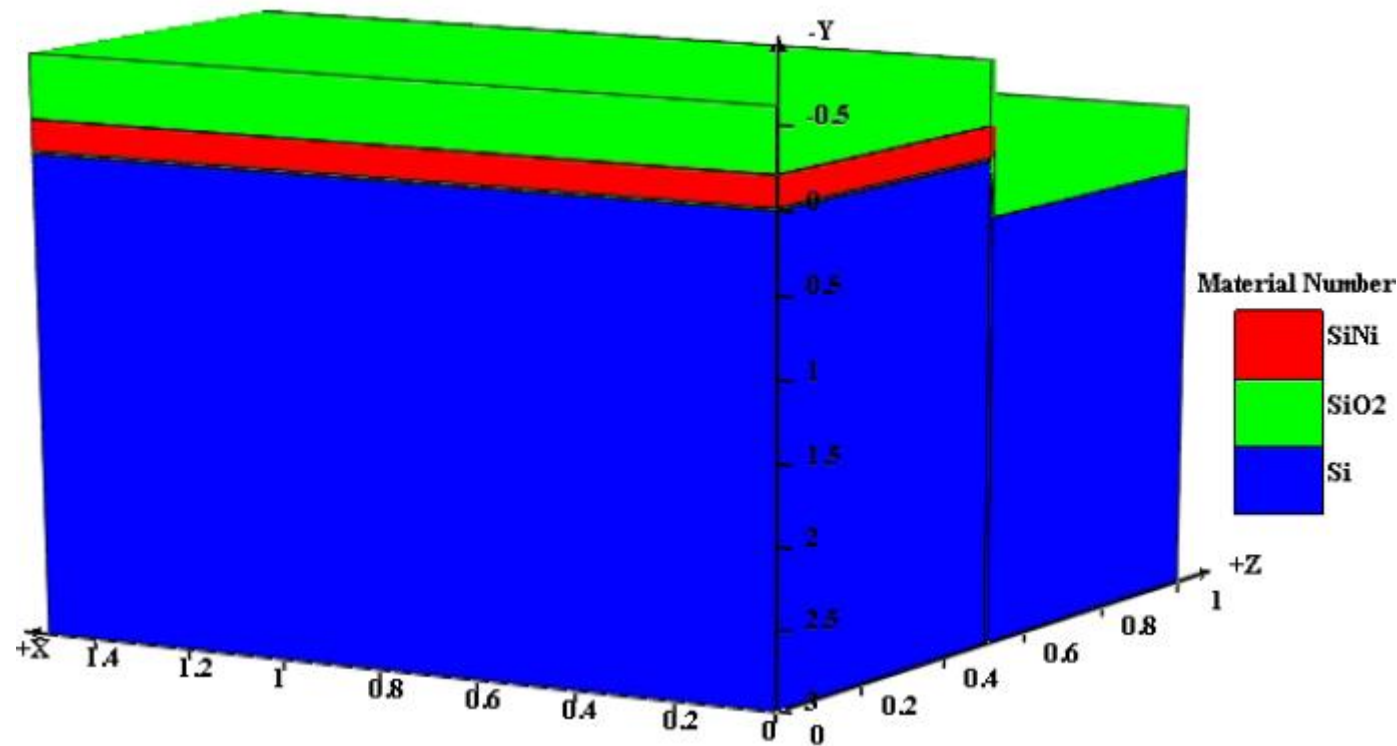
Graphical demonstration

Step 1 STI_04_pullback



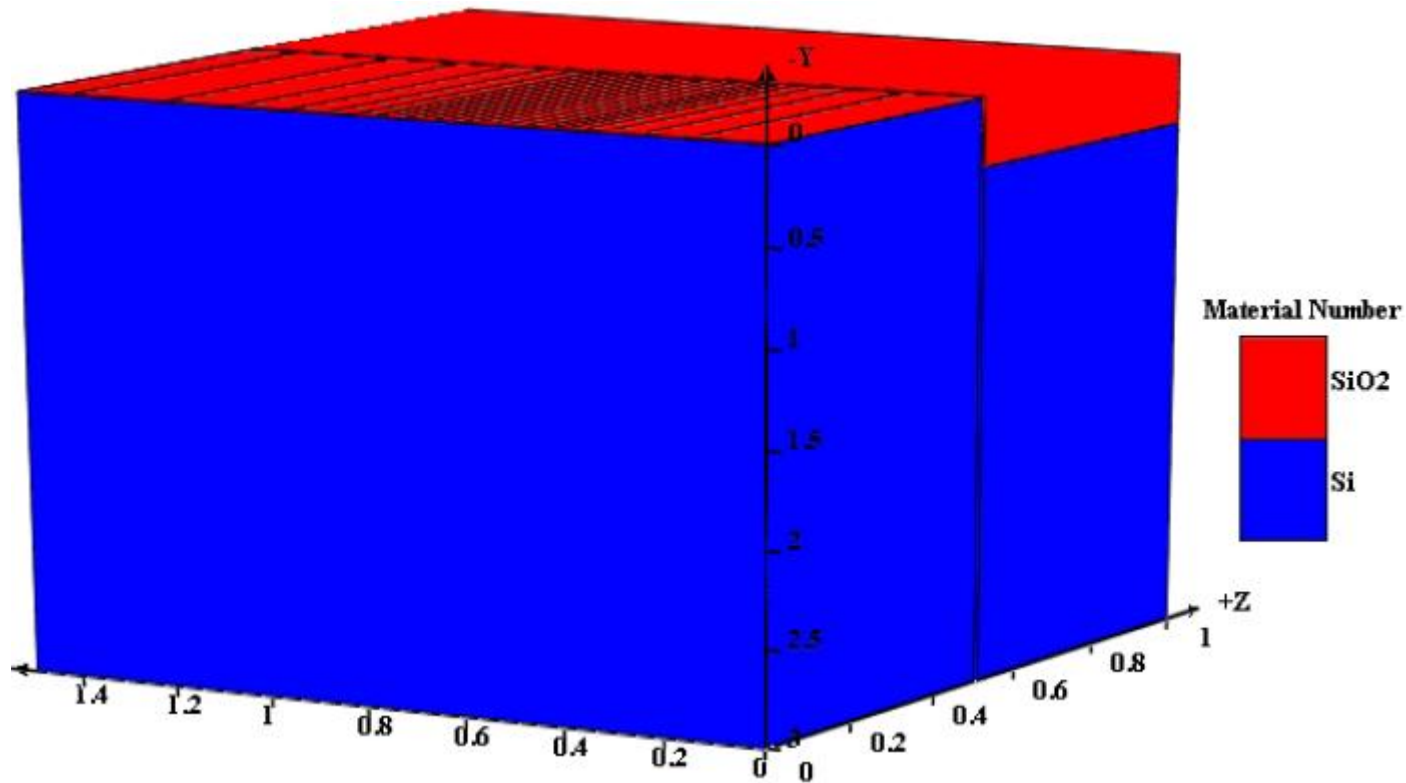
Graphical demonstration

Step 1 STI_05_fill



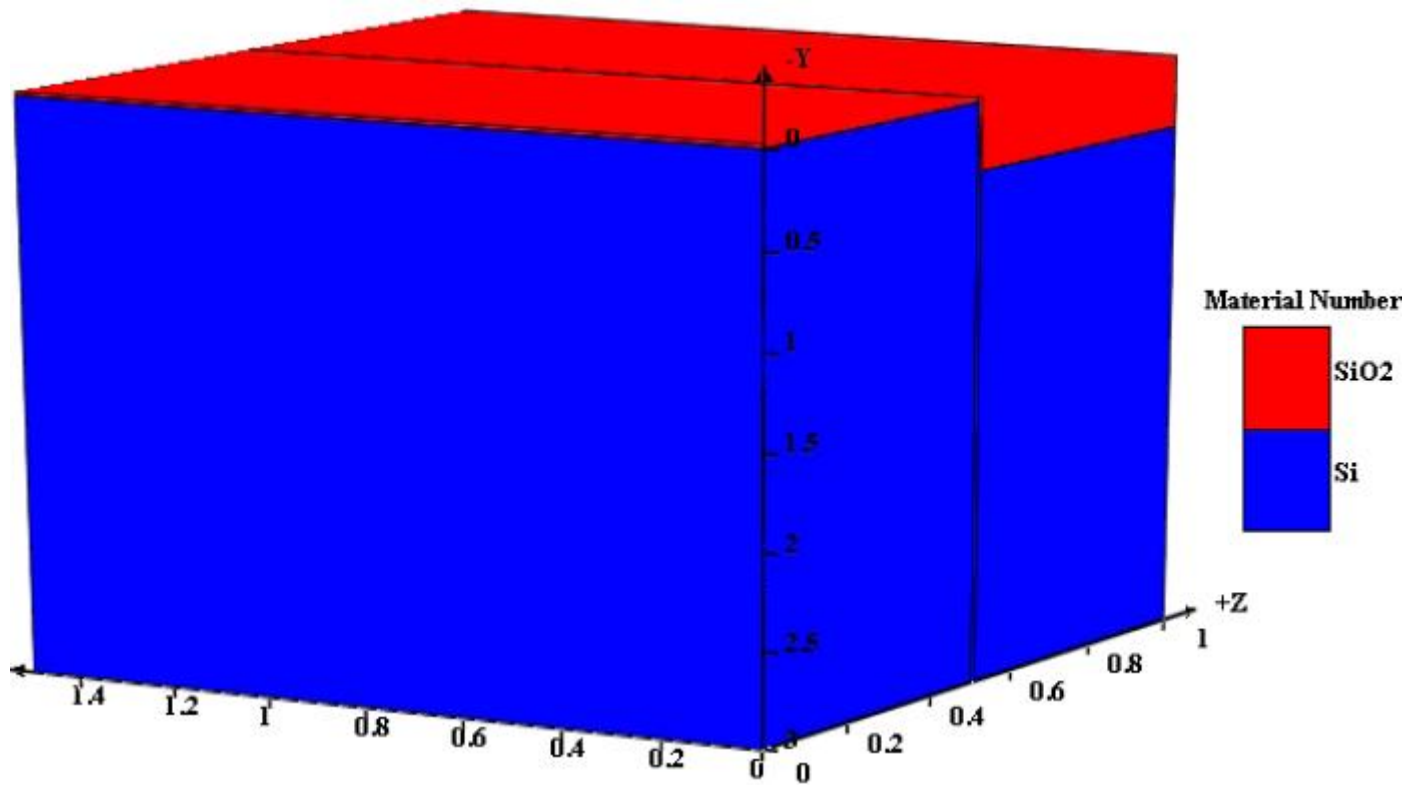
Graphical demonstration

Step 1 STI_06_CMP



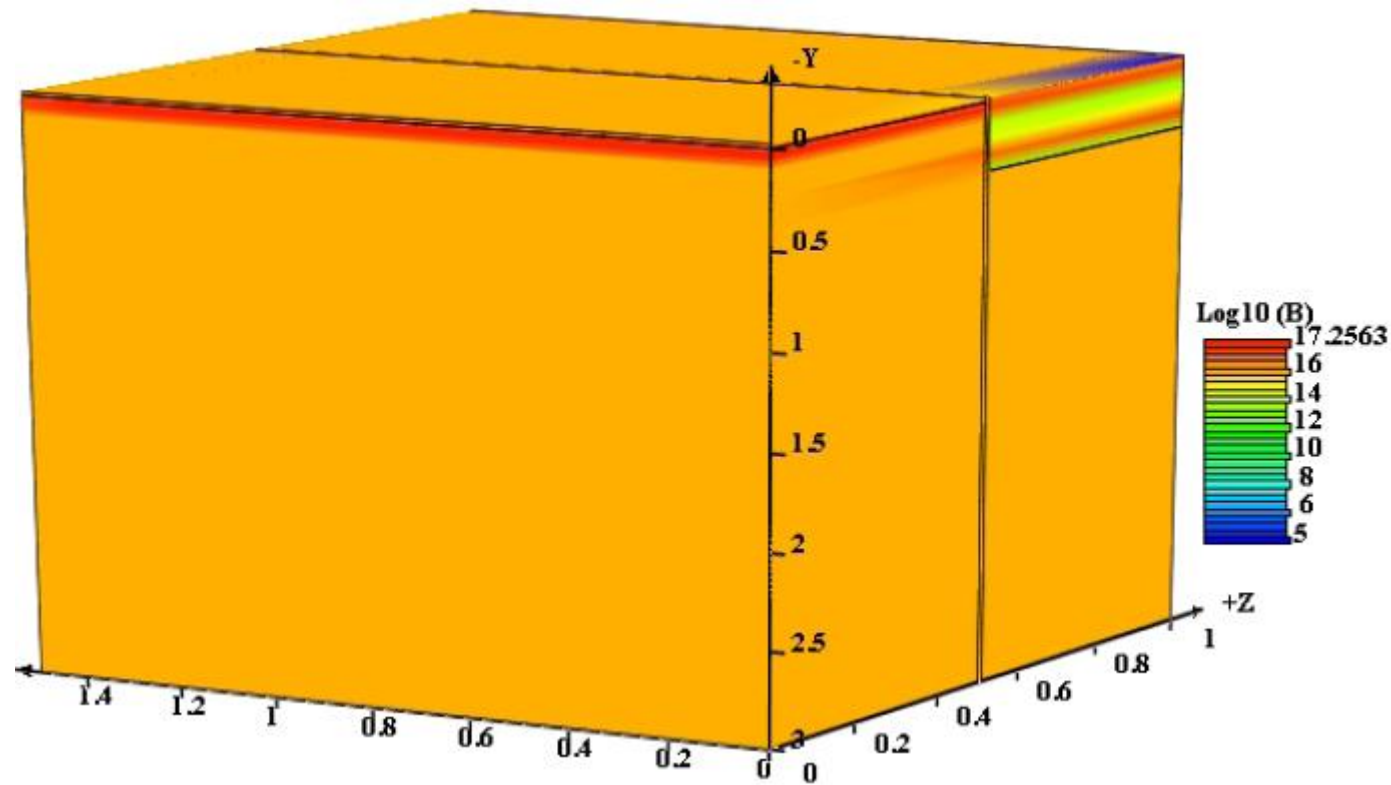
Graphical demonstration

Step 2 gate oxide



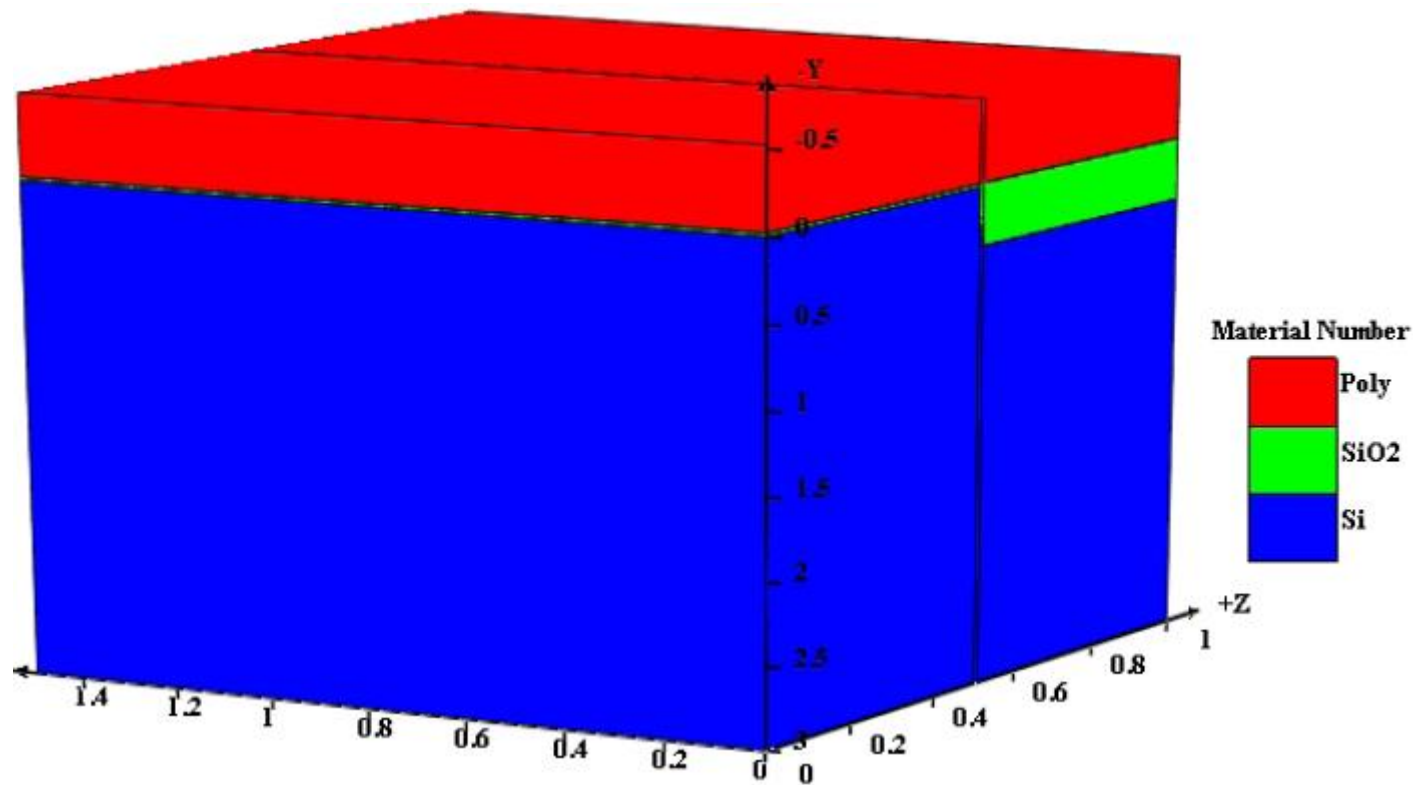
Graphical demonstration

Step 3 channel implant



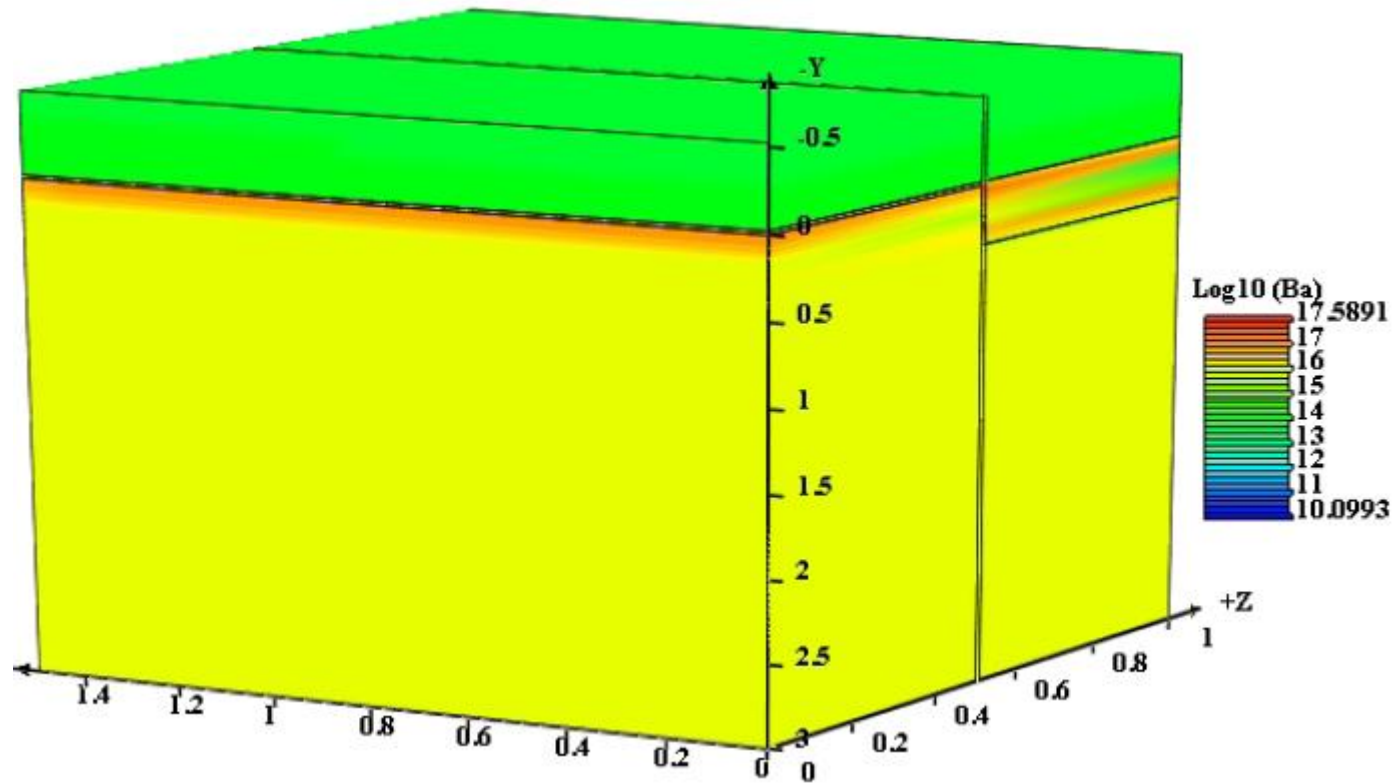
Graphical demonstration

Step 4 poly deposit



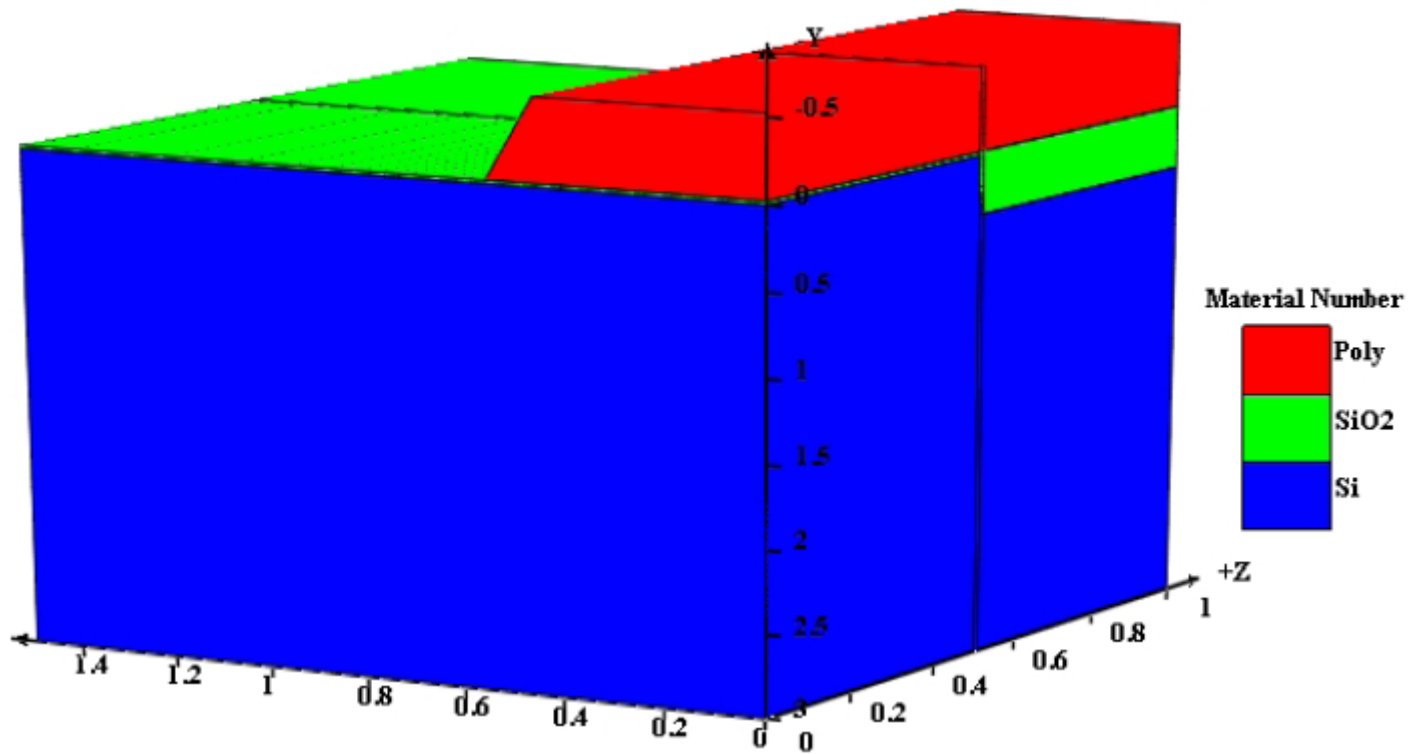
Graphical demonstration

Step 5 poly anneal



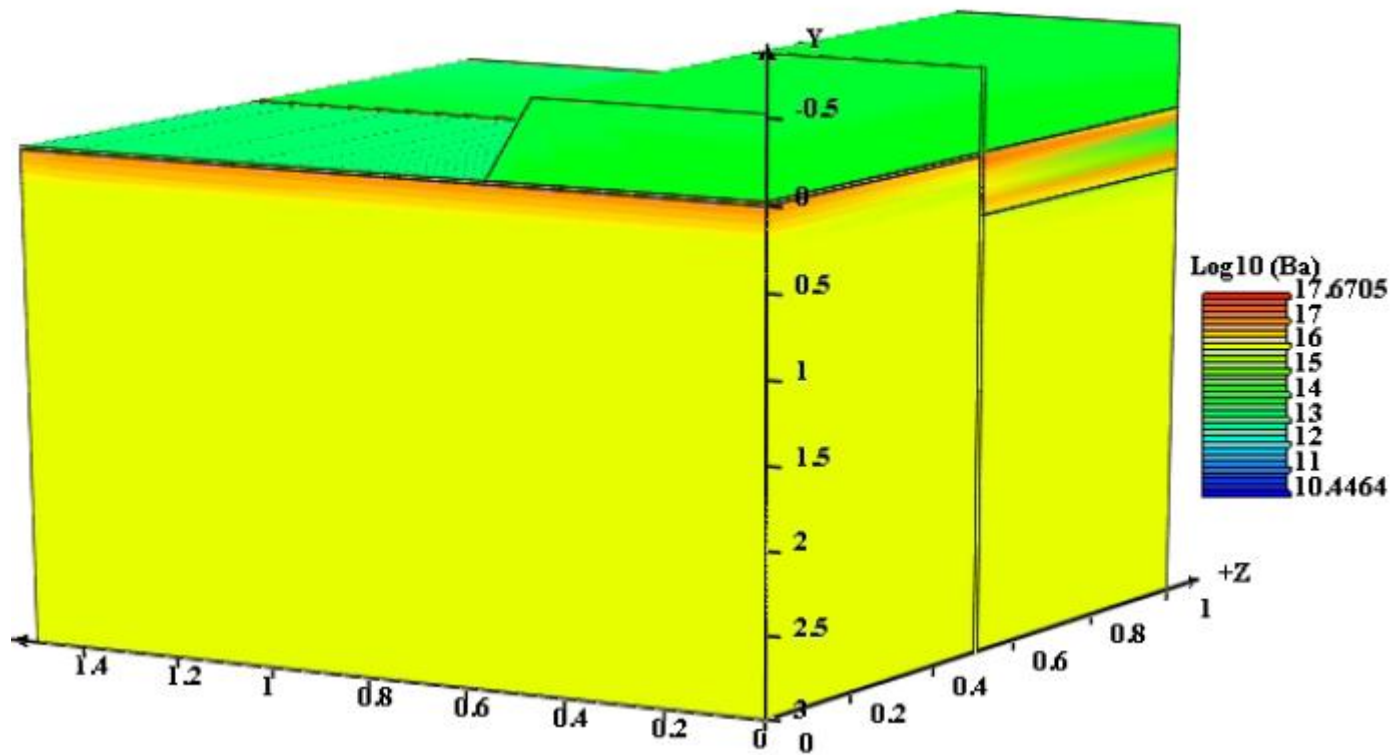
Graphical demonstration

Step 6 poly etch



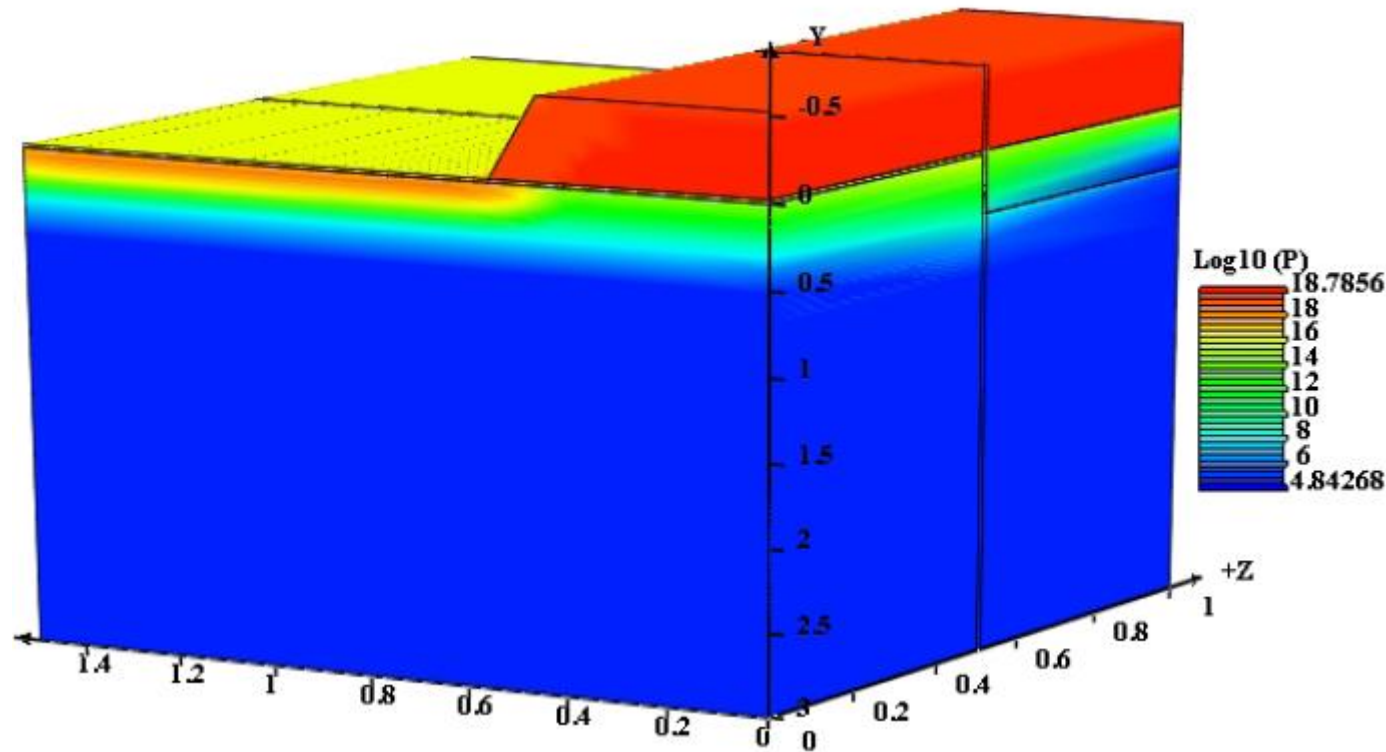
Graphical demonstration

Step 7 anneal after etch



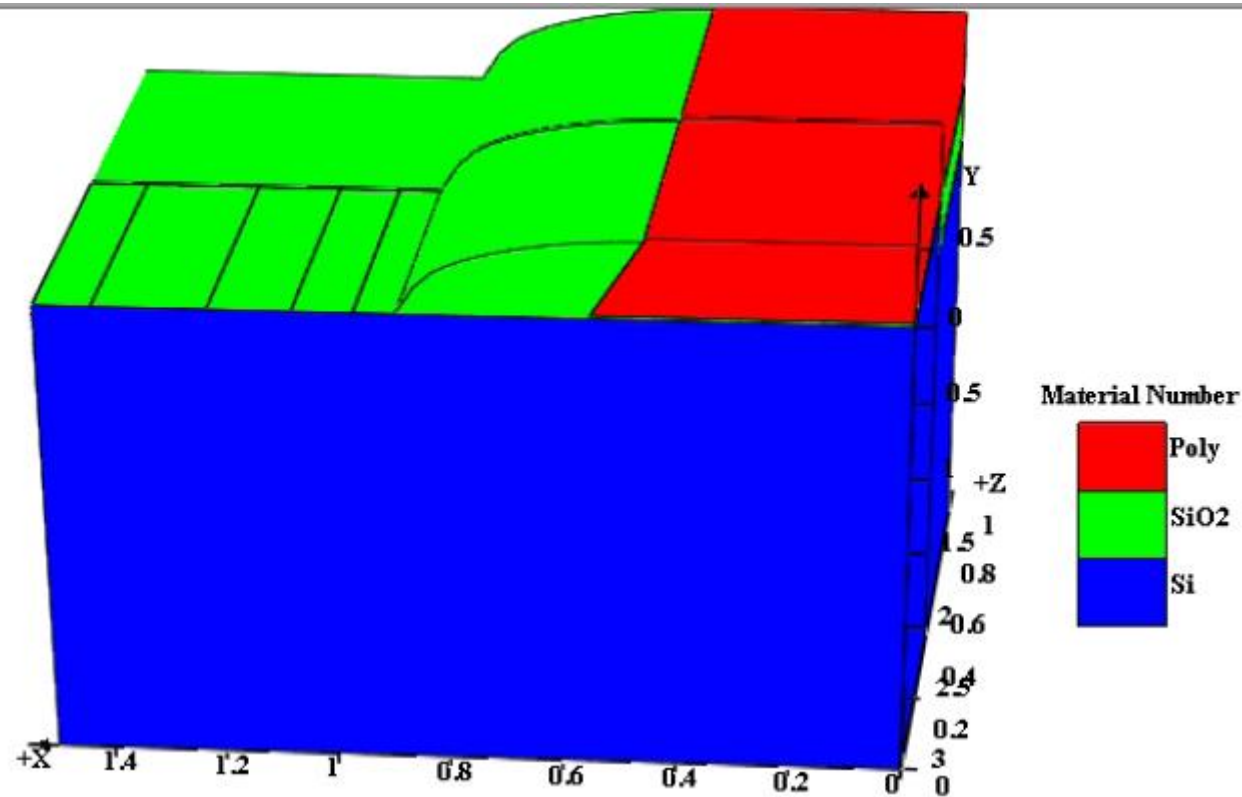
Graphical demonstration

Step 8 Idd implant for source/drain



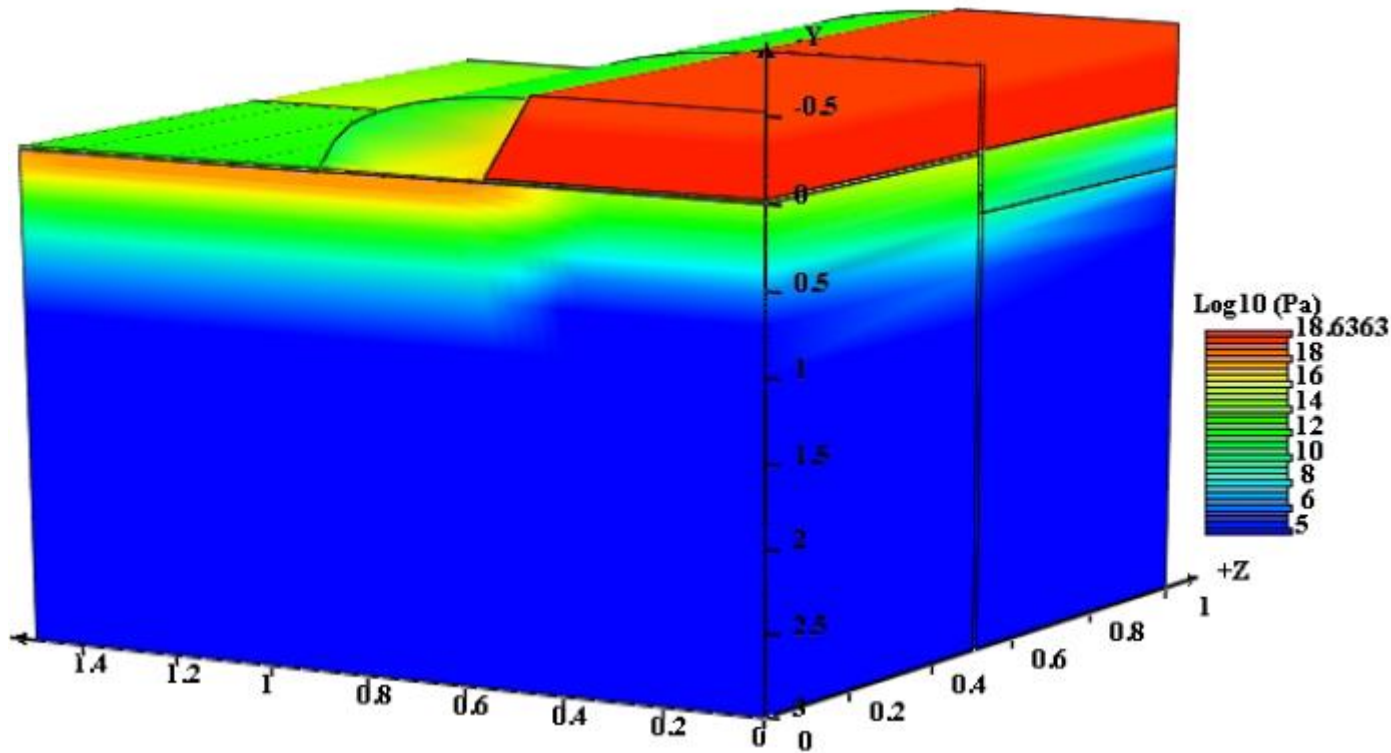
Graphical demonstration

Step 9 make spacer



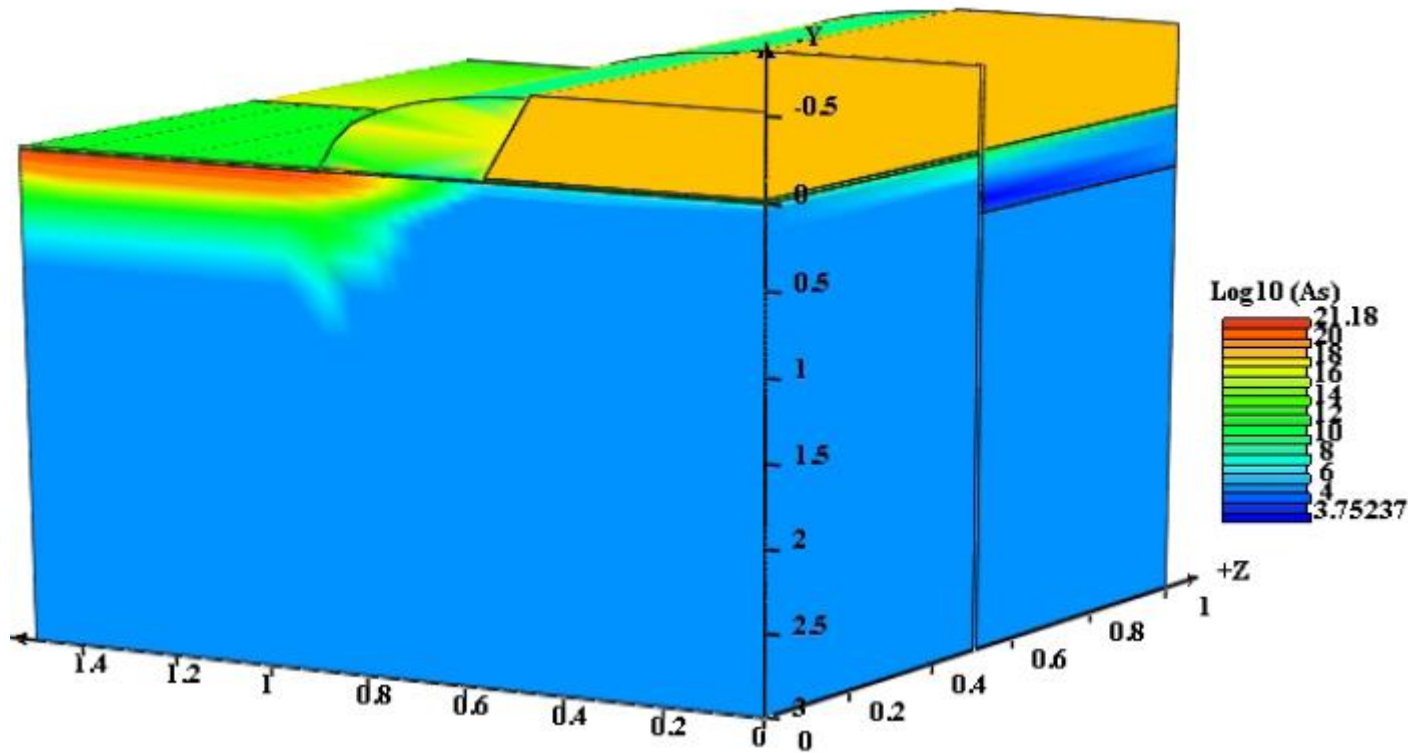
Graphical demonstration

Step 10 anneal in dry O₂ atmosphere



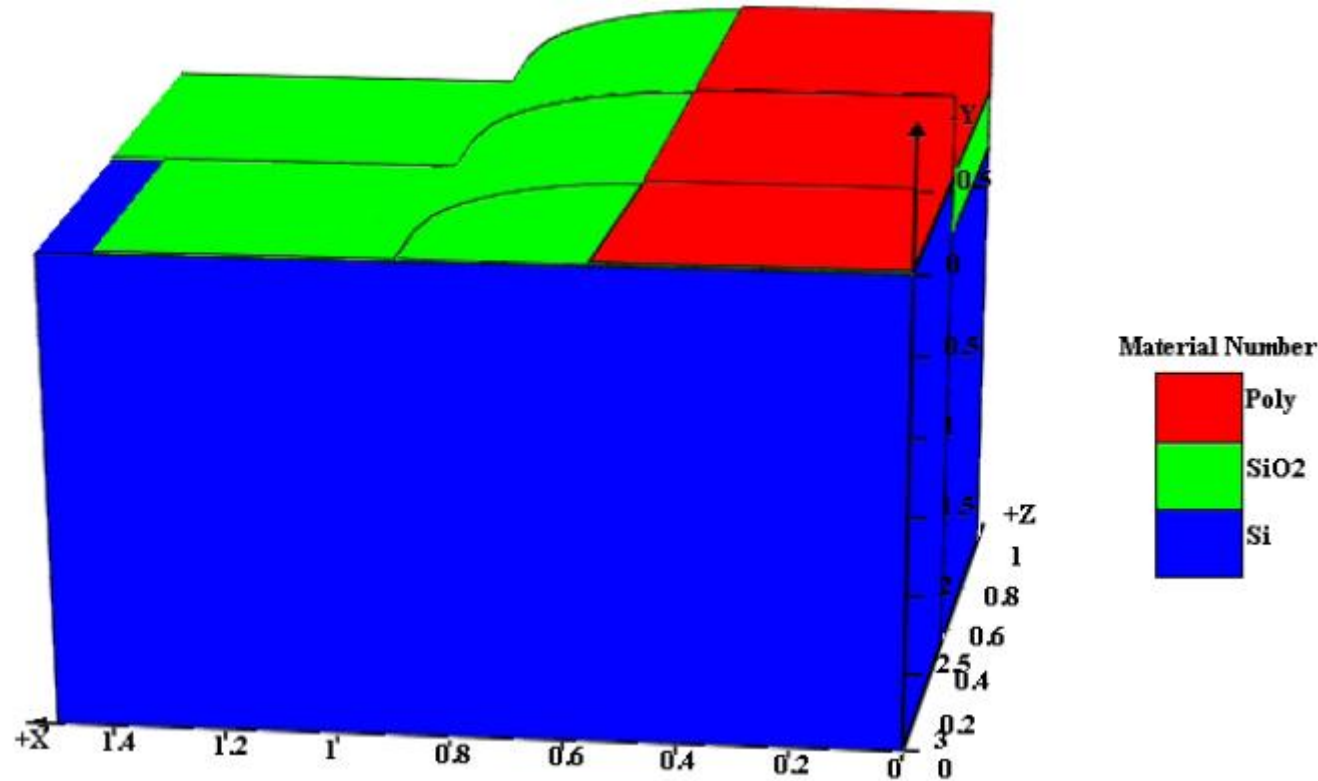
Graphical demonstration

Step 11 final implant for source/drain



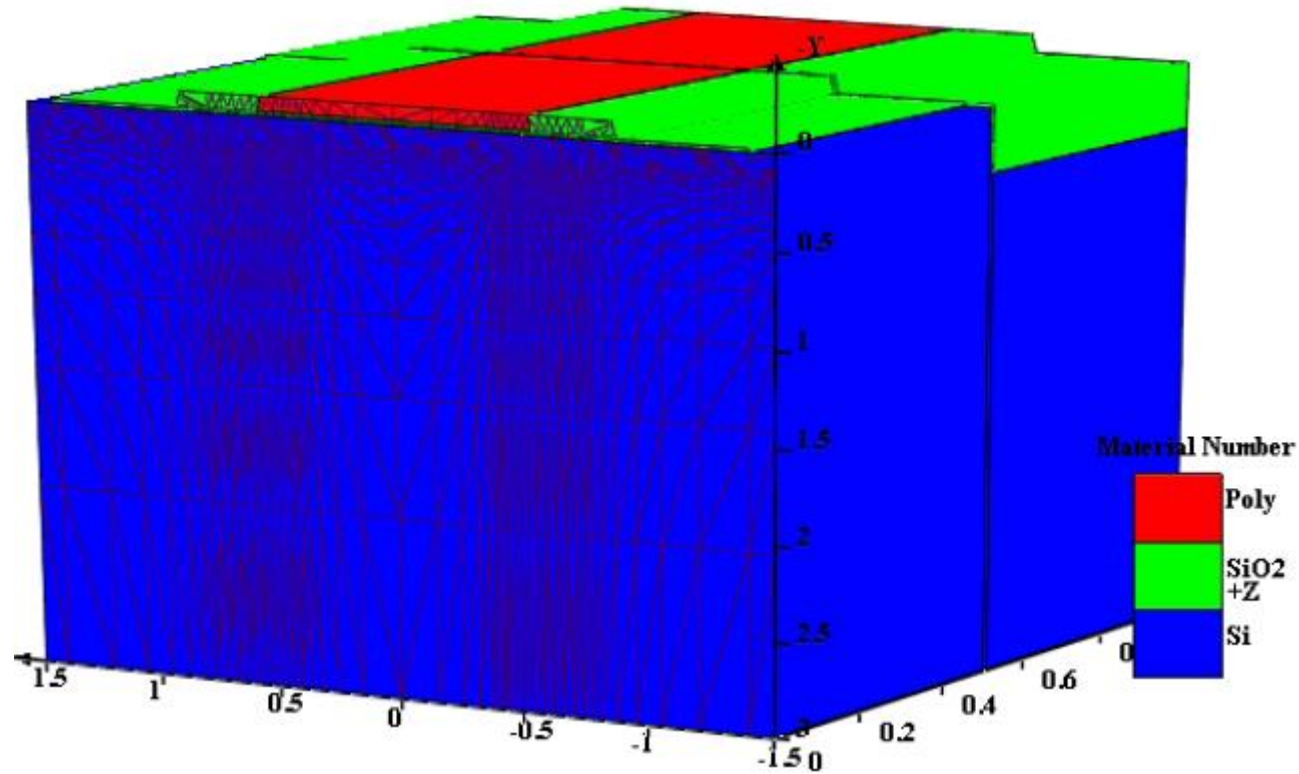
Graphical demonstration

Step 12 etch a window for source/drain contact



Graphical demonstration

Step 13 left mirror the final structure to get a full MOSFET, and some top grids are etched off to save mesh for device simulation.





3D device simulation by Apsys

- ① Define a 3D MOSFET device, including specifications for 3D **contacts**, **materials**, **segments**, etc.
- ② Setup for bias scan
- ③ Plot results

3D device simulation by Apsys

- ① Define a 3D MOSFET device, including specifications for 3D contacts, materials, segments, etc.

\$file:main.sol a line starting with character "\$" is a line for comment
begin indicates that simulation starts here

convention positive_current_flow=inward defines a convention that the positive current direction is that electron flow inward device.

\$ Let us assign contact numbers: 1=source 2=gate 3=drain 4=substrate

define_alias alias=%Lg name=0.5

define_alias alias=%Vd name=voltage_3

define_alias alias=%Id name=current_3

define_alias alias=%Vg name=voltage_2

"define_alias" command is used to define a alias which has a value defined by "name" and will be cited below.

3d_solution_method 3d_flow=yes indicates that this simulation will be a full-3D case.

Part 1 of main.sol

3D device simulation by Apsys

- ① Define a 3D MOSFET device, including specifications for 3D contacts, materials, segments, etc.

```
z_structure uniform_zseg_from=0.0 uniform_zseg_to=0.0 zplanes=1 zseg_num=1  
z_structure uniform_zseg_from=0.5 uniform_zseg_to=0.5 zplanes=1 zseg_num=2  
z_structure uniform_zseg_from=0.51 uniform_zseg_to=0.51 zplanes=1 zseg_num=3  
z_structure uniform_zseg_from=1. uniform_zseg_to=1. zplanes=1 zseg_num=4
```

The upper 4 statements define positions of the x-yplanes at z direction and should be copied from zmesh.szt in CSUPREM project. Also, x-yplanes are numbered in these statements.

load_mesh mesh_inf=sup.aps suprem_import=yes indicates that mesh is imported from a file exported from Csuprem. The imported file name is specified by "mesh_inf"

output sol_outf=sup.out gives a name symbol for output files during simulation.

Part 2 of main.sol

3D device simulation by Apsys

- ① Define a 3D MOSFET device, including specifications for 3D contacts, materials, segments, etc.

`start_loop symbol=%iseg value_from=1 value_to=2`

“start_loop” and “end_loop” define a loop operation. The variable to be looped is specified by “symbol” and its value varies from “value_from” to “value_to” by default step of 1.

`begin_zmater zseg_num=%iseg`

“begin_zmater” and “end_zmater” pack some statements to specify some physical and structural information in the x-y planes numbered in “z_structure” command. Actually, this x-y plane stands for a segment that starts from this x-y plane and ends at the next x-y plane.

`suprem_property silicon_mater=1 oxide_mater=2 poly_mater=3`

“suprem_property” command gives a number for each material which is defined in Csuprem.

`load_macro name=si mater=1`

`load_macro name=sio2 mater=2`

`load_macro name=poly mater=3` defines material name in Apsys software and numbers the material.

[HINT] material names and numbers defined by “load_macro” commands must be accordance with those defined by “suprem_property” command.

Part 3 of main.sol

3D device simulation by Apsys

- ① Define a 3D MOSFET device, including specifications for 3D contacts, materials, segments, etc.

```

impact_chynoweth mater=1
impact_chynoweth mater=3 defines impact ionization rate for material numbered as
"mater".
suprem_contact num=1 xrange=(-1.5 -1.4) side=upper touch_mater=1
suprem_contact num=2 xrange=(-%Lg %Lg) side=upper touch_mater=3
suprem_contact num=3 xrange=(1.4 1.5) side=upper touch_mater=1
suprem_contact num=4 xrange=(-1.5 1.5) side=lower touch_mater=1
"suprem_contact" command specifies position( "side") and size( "xrange") of a
contact for the imported Csuprem structure and gives a number( "num") for this
contact.
contact num=1
contact num=2
contact num=3
contact num=4 defines a contact in Apsys software. Please note that the "num" in
"contact" should be same as that in "suprem_contact"
end_zmater
end_loop

```

Part 4 of main.sol

3D device simulation by Apsys

① Define a 3D MOSFET device, including specifications for 3D contacts, materials, segments, etc.

\$ ----STI segments

start_loop symbol=%isegvalue_from=3 value_to=4

The previous loop specified x-y planes for MOSFET region. This loop specifies x-y planes for STI.

begin_zmater zseg_num=%iseg

suprem_property silicon_mater=1 oxide_mater=2 poly_mater=3

load_macro name=simater=1

load_macro name=sio2 mater=2

load_macro name=poly mater=3

impact_chynoweth mater=1

impact_chynoweth mater=3

suprem_contact num=2 xrange=(-%Lg %Lg) side=upper touch_mater=3

suprem_contact num=4 xrange=(-1.5 1.5) side=lower touch_mater=1

contact num=2

contact num=4

end_zmater

end_loop

Part 5 of main.sol



3D device simulation by Apsys

After physical and structural setup, we will start to calculate device behavior at different bias situation.

3D device simulation by Apsys

② Setup for bias scan

```
newton_par damping_step=5. max_iter=100 print_flag=3
```

“newton_par” command specifies a lot of parameters for Newton iteration which is a mainly mathematical method applied in Apsys software. This command must be put before “equilibrium” and “scan” commands. For detail explain, please refer to its help text and ask technical member in Crosslight.

```
equilibrium outfile_label=biaslab_0
```

“equilibrium” tells simulator calculate physical properties of the device at equilibrium state, including potential, band diagram, etc.

```
newton_par damping_step=1. print_flag=3
```

“newton_par” can be used before different “equilibrium” or “scan” commands to make different settings of Newton iteration.

```
start_loop symbol=%olab value_from=1 value_to=2
```

```
integer_func symbol=%ilab value_from=0 value_to=1
```

```
real_func symbol=%gbias value_from=0.5 value_to=1.0
```

Here, “start_loop” is used to loop a series of “scan” commands. “symbol” variables specified in “start_loop”, “integer_func” and “real_func” commands will be changed simultaneously.

Part 6 of main.sol

3D device simulation by Apsys

② Setup for bias scan

```
scan var=%Vg value_to=%gbias init_step= 0.5000E-01 infile_label=biaslab_%ilab outfile_label=biaslab_%olab
scan var=%Vd value_to= 35.000E+00 scan_label=VDscan_%olab init_step= 0.5000E-01 max_step= 0.4667E+00 &&
  auto_finish=incr_current_3 auto_until=5.e-6 auto_condition=above auto2_finish=voltage_3 auto2_until=5.
auto2_condition=above
```

```
scan var=%Id value_to=250.e-6 print_step=250.e-6 scan_label=IDscan_%olab init_step=0.05e-6 max_step=5.e-6
```

The upper 3 “scan” commands achieves a function that applies a voltage bias(the first “scan”) at MOSFET gate and then applies a voltage(the second “scan”) and current(the third “scan”) bias at MOSFET drain(or source). Bias scan on drain is separated to 2 steps, which is an artifice for mathematical convergence. The second “scan” will be stopped when “auto_finish” condition is satisfied. We can explain the second “scan” as that when current increase of contact 3(“incr_current_3”) is “above” “3.e-6” or when voltage bias of contact 3(“voltage_3”) is “above” “5.” volt, this “scan” will be stopped and the third “scan” will be after this.

The output files from these 3 “scan” commands can be used at post-processing course to get a Vd-Id curve at a specified Vg(“%Vg”). Since the loop loops 2 times here, so we can get 2 Vd-Id curves at different Vg in a family curve plot.

In addition, the unit of Id(“%Id”) is Ampere since this is a 3D case. For 2D cases, Current unit is A/m2. And, “&&” is used to continue a line. For more detail explain, please refer to help text and ask technical member in Crosslight.
end_loop

Part 7 of main.sol

3D device simulation by Apsys

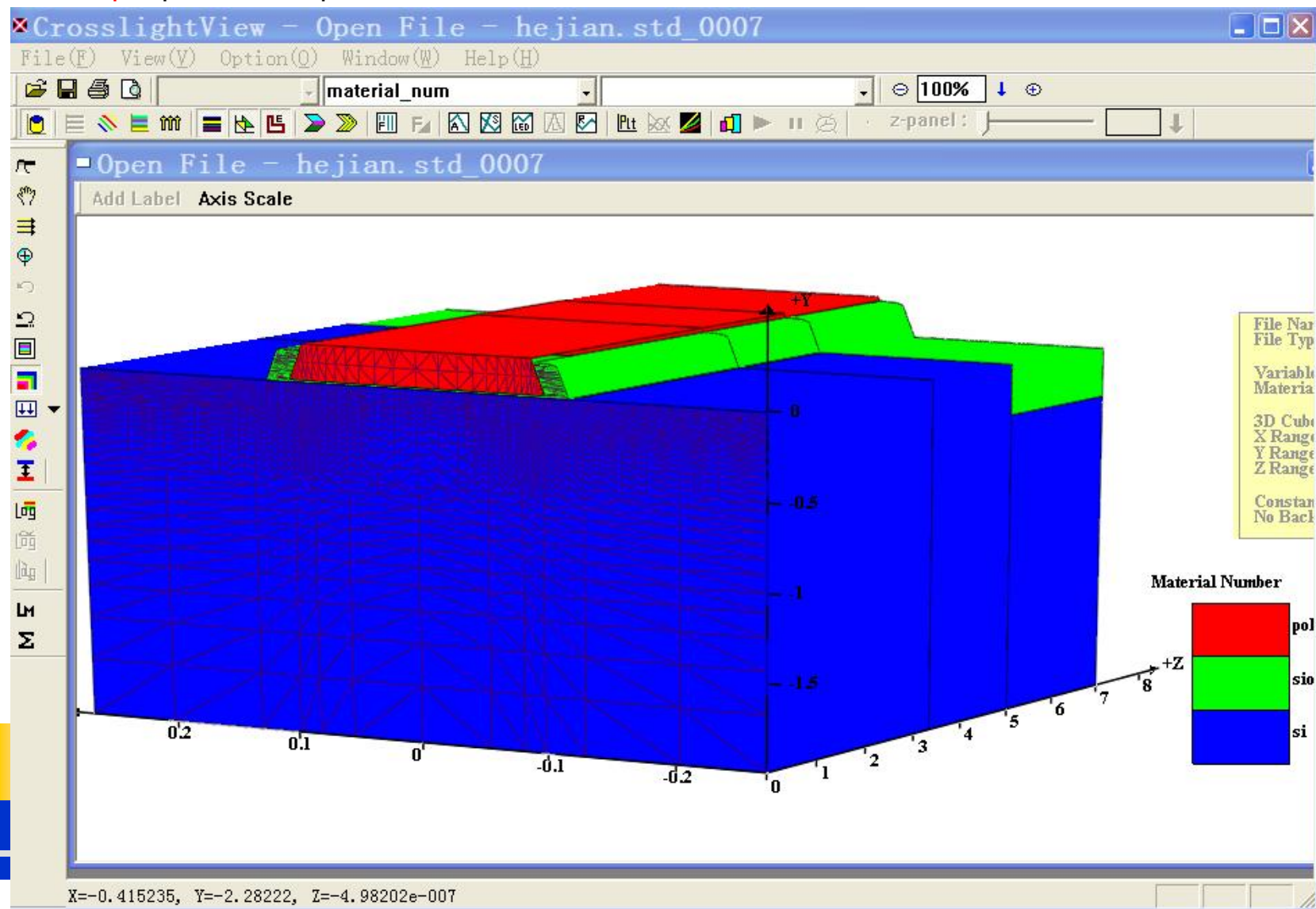
*When simulator finishes device simulation,
we can start to plot some results.*

*Here we use CrosslightView to plot a
family of curves for this MOSFET.*

3D device simulation by Apsys

③ Plot results – curve family

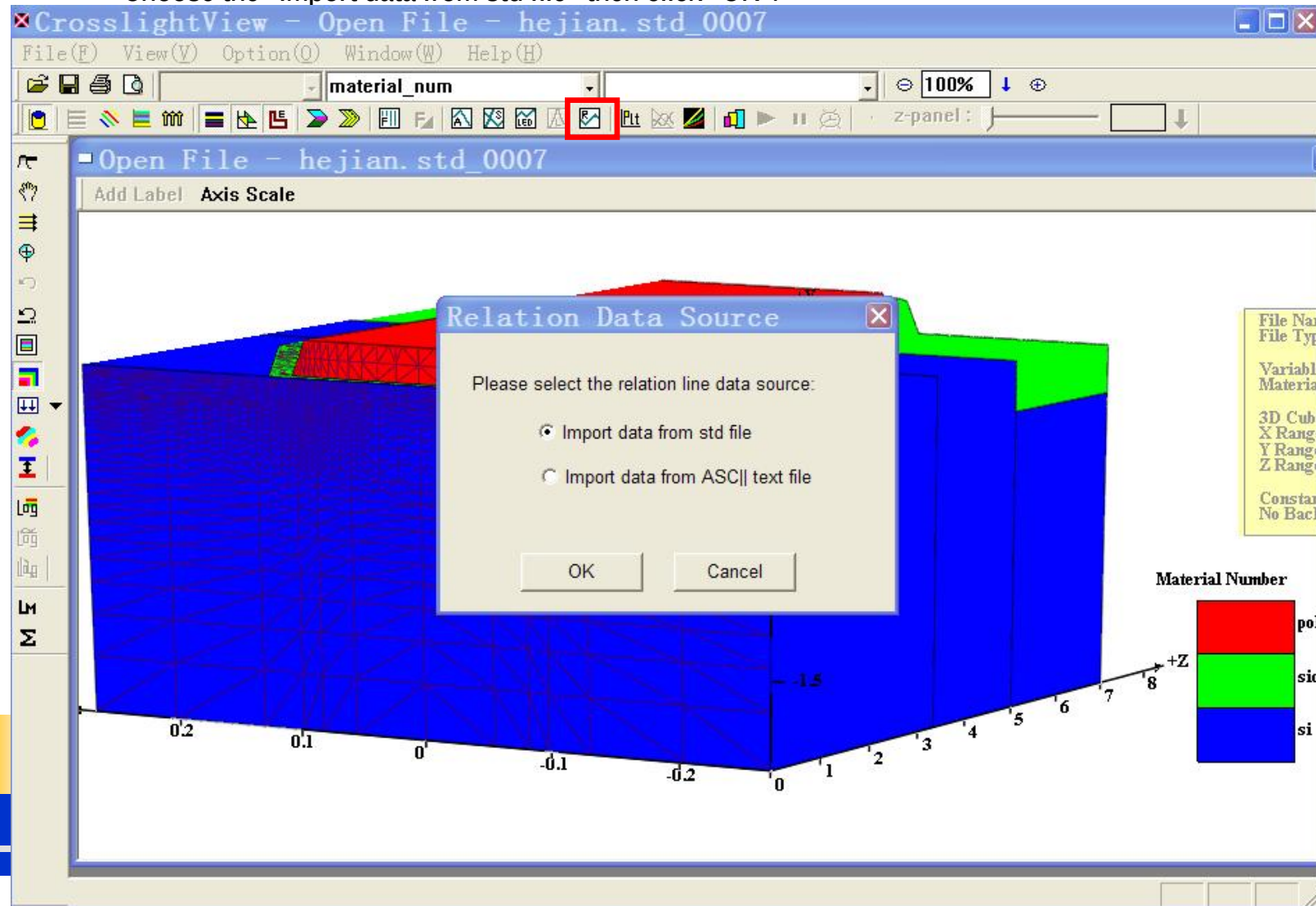
Step1 open a .std output file



3D device simulation by Apsys

③ Plot results – curve family

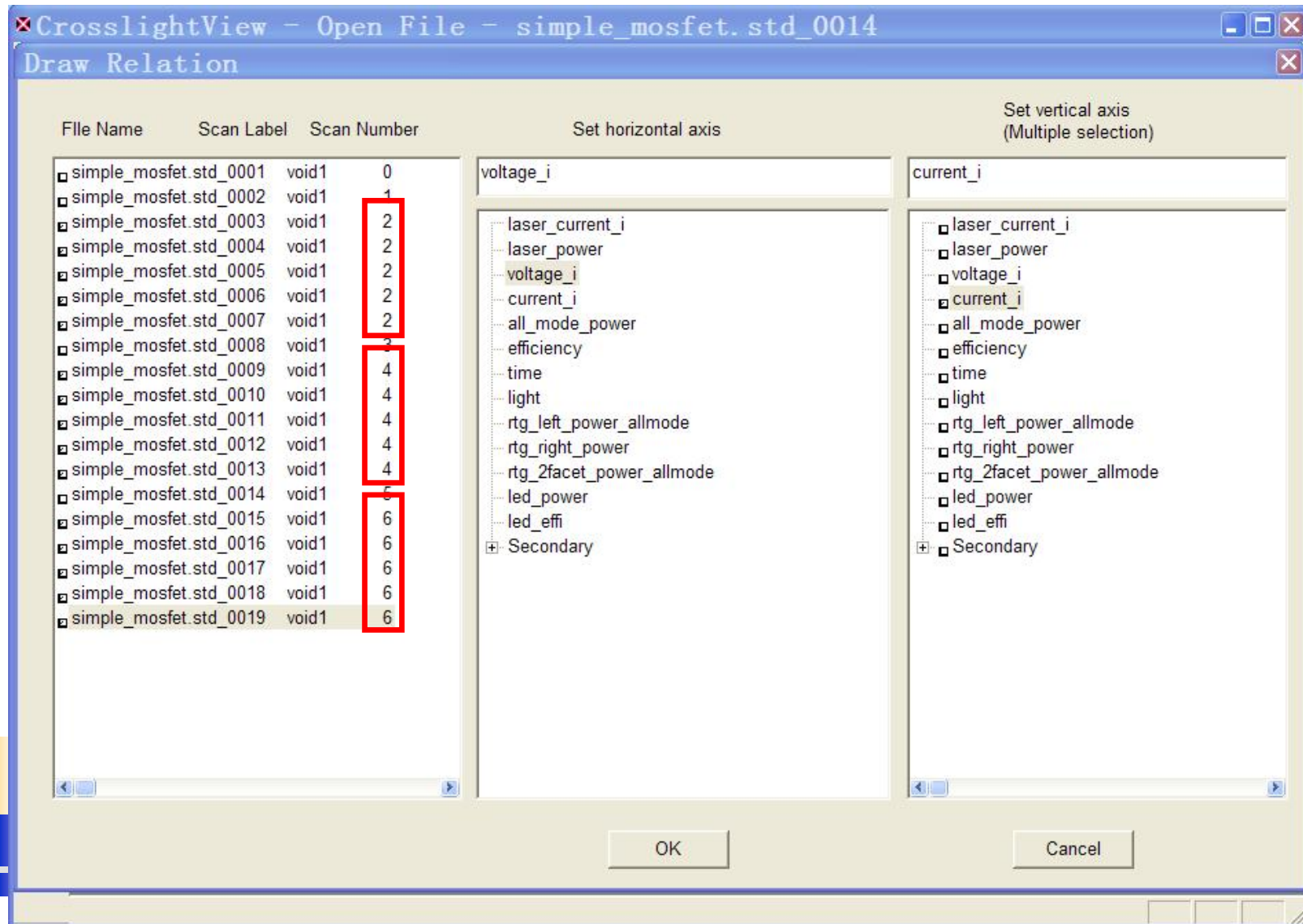
Step2 click the focused icon to select data type which will be used to plot an single **parameter-relation curve**.
Choose the “import data from std file” then click “OK”.



3D device simulation by Apsys

③ Plot results – curve family

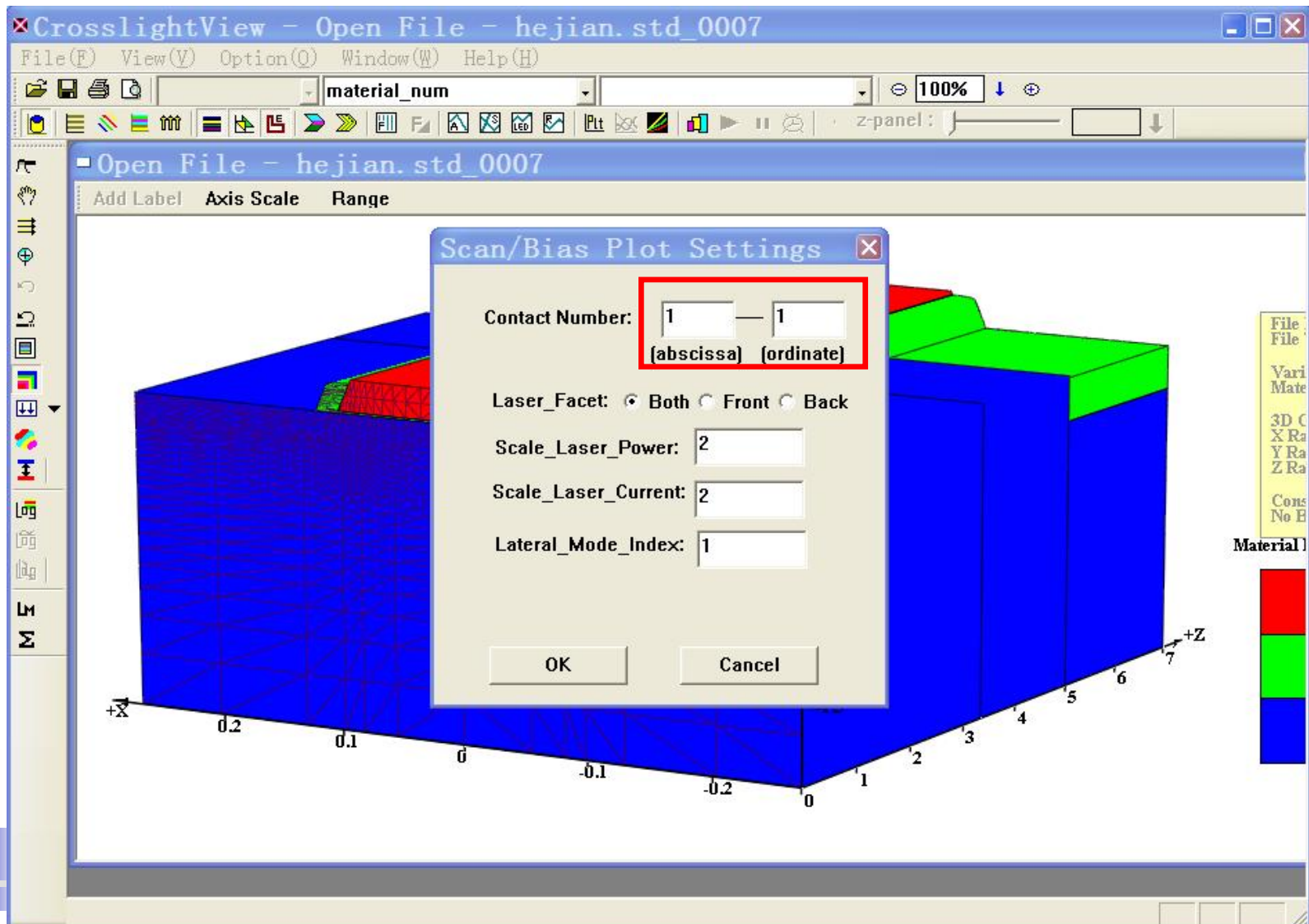
Step3 choose .std file(s) according to “Scan Number”, then choose “voltage_i” and “current_i”, then click “OK”



3D device simulation by Apsys

③ Plot results – curve family

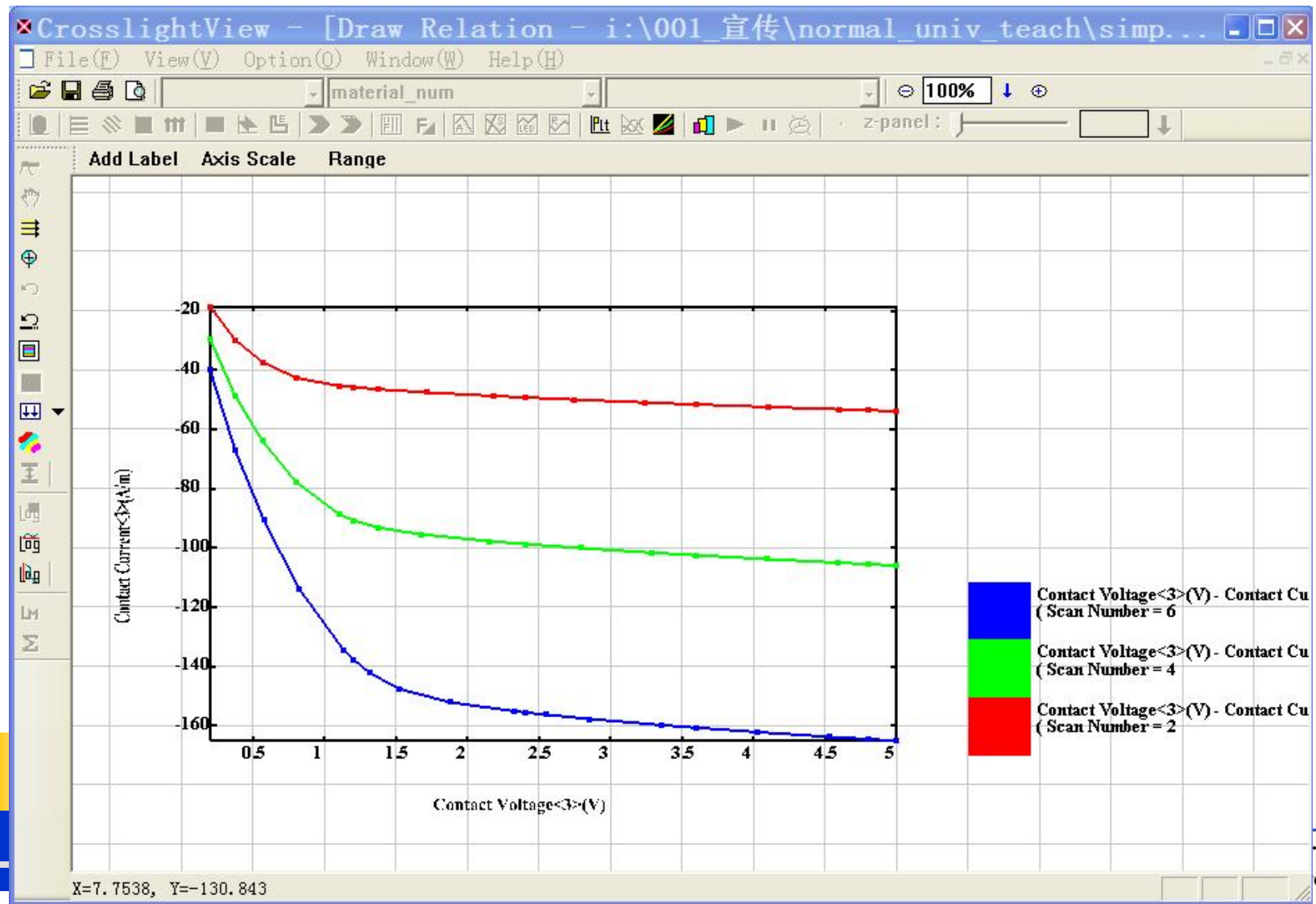
Step4 specify contact number for “i” in “voltage_i” and “current_i”, then click “OK”.



3D device simulation by Apsys

③ Plot results – curve family

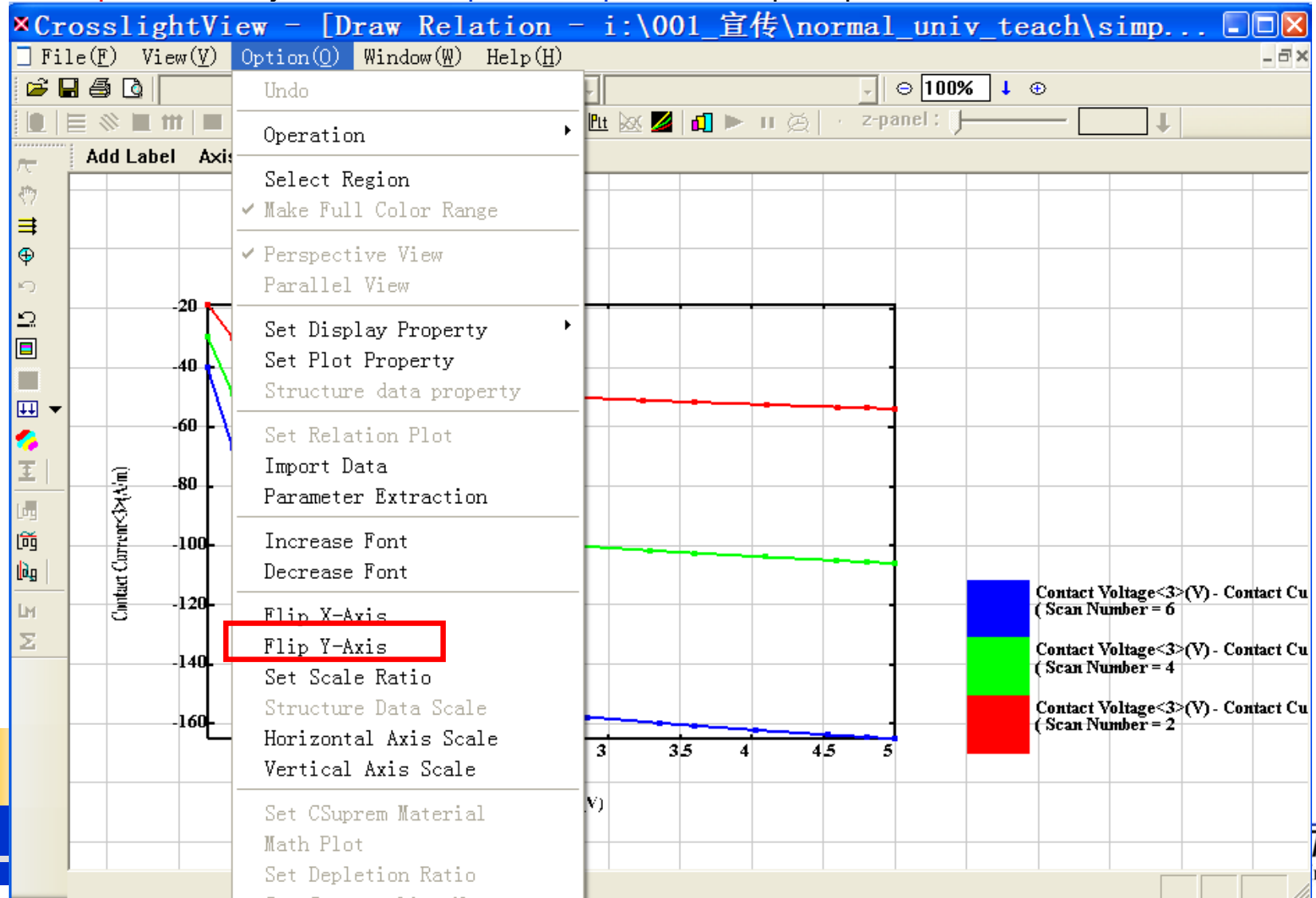
Step5 curve family negative sign relates to the direction of current flow.



3D device simulation by Apsys

③ Plot results – curve family

Step5 curve family Let's click "Option"->"Flip Y-Axis" to flip this plot.



3D device simulation by Apsys

③ Plot results – curve family

Step5 curve family done.

